Blessed G/Michael Catholic Secondary School

Semester 1 Biology Mid1 Examination for Grade11

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade11 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** Which chemical bond links consecutive amino acids together in a polypeptide protein chain?

A. Phosphodiester bond **B.** Glycosidic bond

C. Peptide bond **D.** Ester bond

____ **2.** Enzymes accelerate biochemical reactions primarily by:

A. Increasing temperature **B.** Lowering activation energy

C. Changing the equilibrium constant **D.** Supplying ATP directly

____ **3.** Which model of enzyme action proposed that the active site undergoes conformational changes upon substrate binding?

A. Lock-and-key model **B.** Induced-fit model

C. Fluid mosaic model **D.** Sliding filament model

____ **4.** A competitive enzyme inhibitor binds to:

A. The active site directly, competing with the substrate
B. An allosteric site only
C. The enzyme-substrate complex only
D. The product molecules

____ **5.** Which disaccharide sugar is composed of glucose and fructose linked by a glycosidic bond?

A. Maltose **B.** Lactose **C.** Sucrose **D.** Cellulose

____ **6.** Which class of biomolecules has the primary cellular role of storing and transmitting genetic information?

A. Triglycerides **B.** Polypeptides

C. Nucleic acids (DNA and RNA) **D.** Phospholipids

____ **7.** Which test reagent produces a purple-violet color in the presence of peptide bonds in proteins?

A. Benedict’s reagent **B.** Iodine solution

C. Biuret reagent **D.** Sudan III stain

____ **8.** Which of the following is an example of a structural polysaccharide in plants?

A. Cellulose **B.** Glycogen **C.** Starch **D.** Sucrose

____ **9.** Triglycerides are formed through ester condensation reactions between one glycerol molecule and:

A. Three fatty acid chains **B.** Two amino acids

C. Three monosaccharides **D.** One phosphate group

____ **10.** What is the primary structure of a protein?

A. The linear sequence of amino acids linked by peptide bonds
B. Alpha-helix and beta-pleated sheet folding
C. Three-dimensional globular subunit arrangement
D. Interaction of multiple polypeptide chains

____ **11.** How do temperatures far above an enzyme's thermal optimum cause loss of catalytic function?

A. By breaking hydrogen and ionic bonds, denaturing tertiary protein structure
B. By hydrolyzing peptide bonds into free amino acids
C. By lowering substrate concentration
D. By converting the active site into RNA

____ **12.** In non-competitive enzyme inhibition, the inhibitor binds to:

A. An allosteric site other than the active site
B. The substrate molecule directly
C. The active site competitively
D. The enzyme product

____ **13.** What is the Michaelis constant (Km) of an enzyme-catalyzed reaction?

A. The substrate concentration at which the reaction velocity is half of Vmax
B. The maximum reaction rate achievable
C. The optimum temperature in Celsius
D. The turnover number kcat

____ **14.** Which inorganic ion acts as an essential cofactor for salivary amylase?

A. Chloride ion (Cl-)
B. Iron ion (Fe2+)
C. Magnesium ion (Mg2+)
D. Copper ion (Cu2+)

____ **15.** Which of the following is a purine nitrogenous base found in both DNA and RNA?

A. Adenine **B.** Cytosine **C.** Uracil **D.** Thymine

____ **16.** Phospholipids are described as amphipathic molecules because they contain:

A. A hydrophilic phosphate head and two hydrophobic fatty acid tails
B. Two hydrophilic heads and no fatty acids
C. Only non-polar steroid rings
D. Four glycerol backbones

____ **17.** Which level of protein organization features the association of multiple folded polypeptide subunits into a single multi-protein complex (e.g. hemoglobin)?

A. Quaternary structure
B. Tertiary structure
C. Secondary structure
D. Primary sequence

____ **18.** During feedback inhibition in metabolic pathways, the end product typically inhibits:

A. The first committed enzyme of the metabolic pathway
B. The final enzyme in the pathway
C. All enzymes equally
D. The DNA promoter region

____ **19.** Which test identifies reducing sugars like glucose and maltose by forming an orange-red cuprous oxide precipitate upon heating?

A. Benedict's test
B. Iodine test
C. Biuret test
D. Emulsion test

____ **20.** What type of chemical reaction links monosaccharides into polysaccharides while eliminating water?

A. Condensation (dehydration synthesis)
B. Hydrolysis
C. Oxidation
D. Decarboxylation

____ **21.** Which cellular organelle packages proteins into membrane-bound transport vesicles for secretion?

A. Golgi apparatus
B. Lysosome
C. Smooth ER
D. Peroxisome

____ **22.** In aerobic cellular respiration, where does the Krebs (citric acid) cycle take place within the eukaryotic cell?

A. Mitochondrial matrix
B. Inner mitochondrial cristae membrane
C. Cytoplasm cytosol
D. Nuclear envelope

____ **23.** Which nucleic acid molecule transfers specific amino acids to ribosomes during mRNA translation?

A. tRNA (transfer RNA)
B. rRNA (ribosomal RNA)
C. mRNA (messenger RNA)
D. snRNA (small nuclear RNA)

____ **24.** What is the storage polysaccharide stored in human liver and skeletal muscle cells?

A. Glycogen **B.** Cellulose **C.** Amylose **D.** Chitin

____ **25.** Prokaryotic bacterial ribosomes have a sedimentation coefficient of:

A. 70S (composed of 50S and 30S subunits)
B. 80S (composed of 60S and 40S subunits)
C. 60S
D. 100S